Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RA52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

AUTOMATION IN PROCESS CONTROL

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Define Programmable logic controllers. Describe the major components of PLCs with a suitable block diagram.	**07**
	b.	List the different PLC modules used in an automation system and explain the chassis/Rack and Central processing units are used in PLC.	**07**
2.	**a.**	Explain the use of two internal relays and starting of multiple outputs using a ladder diagram.	**06**
	b.	Construct the different types of Timers and explain the motor sequence using ladder logic programming.	**08**
3.	**a.**	Show how the data transfer instructions are formed from one word or register to another and explain the move instruction used to change the preset time of a timer using ladder logic programming.	**10**
	b.	Discuss the following word-level math instructions **i.** NEQ **ii.** FRD	**04**

OR

4.	**a.**	Explain the different types of bit shift instructions using ladder function.	**06**
	b.	List and explain the commands in math instructions. Discuss the operation of vessel overfull alarm program.	**08**
5.	**a.**	Construct and explain the process of centralised and decentralised computer control concept.	**08**
	b.	Explain the continuous process displays in process and plant information with a suitable diagram.	**06**

OR

6.	**a.**	Discuss the different levels of distributed control system with the help of a diagram.	**08**
	b.	Construct the Leeds and Northup Max-I system and mention the basic components of system for the same.	**06**

7.	a.	Show the FOUNDATION Fieldbus architecture and discuss the different types of blocks in communication layer.	07
	b.	Illustrate and explain the remote terminal units. Mention its applications.	07
		OR	
8.	a.	Sketch and explain the architecture of Dual Modular Redundancy in Reliable system development strategy	07
	b.	Develop the flow diagram of General approach to model building and explain the same.	07